

Matej Ciglenečki

CTO with over 4 years of experience, building a distributed machine learning system serving tens of thousands of active users. **Actively preparing to pursue a machine learning PhD, with a primary goal of publishing in top-tier ML conferences.** I'm a strong communicator who can present findings clearly and understandably. Throughout my experience, I've developed excellent research and problem-solving capabilities accompanied by a strong sense of ownership. My research interests include video understanding, image editing, and 3D/4D reconstruction, with openness to related areas.

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Education

M.Sc Data Science – Faculty of Electrical Engineering and Computing , Zagreb, Croatia master's thesis: Generative Novel View Synthesis Using Neural Radiance Fields (NeRF) relevant courses: <i>Advanced Algorithms and Data Structures, Machine Learning, Multivariate Data Analysis, Deep learning, Parallel Programming, Signal processing, Natural Language Processing, Numerical Mathematics, Deep Generative Modeling</i>	Oct 2021 – July 2023
B.Sc Computer Science – Faculty of Electrical Engineering and Computing , Zagreb, Croatia bachelor's thesis: Facial Expression Recognition Using Convolutional Neural Networks relevant courses: <i>Algorithms and Data Structures, OOP, Design Patterns, Databases, Probability and Statistics, Artificial intelligence</i>	Oct 2017 – July 2021

Experience

TensorPix – CTO • Trained a super resolution model conditioned on the degradation embedding that's currently in production (PyTorch). • Built and trained a model that predicts an image's degradation embedding (blur, compression, noise...) (PyTorch). • Built a high-quality image filtering pipeline to filter training data, improving the super resolution model quality metrics. • Developed a custom video inference engine for efficient video processing (PyTorch, Python, ONNX, TensorRT). • Researched and benchmarked and torch.compile for the purpose of inference cold start with static and dynamic input shapes. • Enhanced generative modeling benchmarking and evaluation suite (PyTorch, Python). • Managed 3 engineers. Heavily influencing the company's strategy. Pitched to investors during Series A fundraising.	Dec 2024 – present
TensorPix – Machine Learning Engineer • Researched, implemented, and trained models for super resolution and low-light video enhancement (PyTorch, Python). • Implemented a task queue and distributed workers for ML inference used in production (Redis, Python, Django). • Implemented a custom cloud-agnostic instance auto scaler (Redis, Python).	Nov 2023 – Dec 2024
Microblink – Machine Learning Engineer Intern • Researched and implemented NeRF to support the generation of segmentation masks via transfer learning. (PyTorch) • Developed NeRF scene cropping method to isolate the central object to reduce foreground image artifacts. (PyTorch) • Processed point clouds and images, numerically validated view extrapolation, mesh reconstruction, generalization across different scenes and correlation between scene's variability and reconstruction quality (NeRF, Python, PyTorch). • Researched and implemented a font classifier via semi-supervised learning used for document verification (PyTorch).	Feb 2023 – Sep 2023
Photomath (Google) – Software Engineer Intern • Developed cloud services that parse, transform, enrich and deliver millions of events from Photomath's large textbook solutions database (Python, GCP, Datastream, Pub/Sub, Dataflow, Cloud Run,s GitHub Actions, Spark).	July 2022 – Oct 2022
Memgraph – Software Engineer • Designed and implemented the database schema and backend for Memgraph Cloud. It manages AWS EC2 instances, and supports monthly user billing based on user usage (node.js, TypeScript, Express, Sequelize, PostgreSQL). • Set up observability Elastic Stack on AWS EC2 to analyze application logs via AWS CloudWatch in Kibana dashboards.	July 2020 – Oct 2021

Skills

Languages:	Python, TypeScript, bash, C++
Technologies:	git, Linux, PyTorch, OpenCV, numpy, Docker, LaTeX
Other:	Machine Learning, Computer Vision, Generative Models, Distributed Systems, NLP

Publications

A Squeeze of LIME, a Pinch of SHAP: Adding the Flavor of Explainability to Sentence-Level Commonsense Validation (M. Moslavac, R. Grbelja, M. Ciglencečki), <i>Text Analysis and Retrieval 2023 Course Project Reports</i> • Trained a DeBERTa-based model for classification of sentence coherence. • Analyzed explainability of the trained model with LIME and SHAP, model-agnostic and post-hoc methods.	
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Extracurricular

Machine Learning competition – Musical Instrument Classification • Led a team of four. Won the 2nd place and 1st place in model performance. • Trained a model that labels 11 musical instruments from an input audio signal, developed in PyTorch using deep learning, digital signal processing, audio feature engineering, and computer vision style spectrogram/image representations derived from raw audio. Deployed the trained model behind a FastAPI inference service. • Wrote both competition report and technical documentation. • LUMEN Data Science is the largest machine learning competition in the Croatia.	2023
A Squeeze of LIME, a Pinch of SHAP – an NLP paper • Wrote and published an NLP paper with two other authors. • Trained a DeBERTa-based model for classification of sentence coherence. • Analyzed explainability of the trained model with LIME and SHAP, model-agnostic and post-hoc methods.	2023
Machine Learning competition – Computer Vision GeoGuessr • Led a finalist team of three. Won the 2nd place in model performance. • Trained a model that predicts the lat/lng location of Google Street View images, developed in PyTorch using deep learning, computer vision, and geospatial feature engineering. Deployed the trained model behind a FastAPI inference service. • Wrote both competition report and technical documentation. • LUMEN Data Science is the largest machine learning competition in the Croatia.	2022
Paper implementation – driver fatigue detection in an EEG-based system • Reproduced paper's baseline and methods. Trained the model that matches the reported performance. • Analyzed and extracted entropy features from driver's EEG data. • Trained and compared four different models, achieving +1% higher performance than reported in the paper.	2022

Other

Certificate in Advanced Rhetoric and Public Speaking (Heureka)	2026
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Attended Croatian Machine Learning Workshop (CMLW)	2024
Certificate in Soft Skills (EESTEC)	2019